

READING PASSAGE 2

The Melting of Arctic Sea Ice

A The Melting of Arctic Sea Ice has emerged as one of the most significant topics in contemporary academic discourse. The complexity and multifaceted nature of this subject have attracted the attention of researchers, policymakers, and practitioners across multiple disciplines, all seeking to understand its implications for the future.

B The origins of contemporary research in this area can be traced to pioneering studies conducted during the mid-twentieth century. These early investigations, though constrained by the technological and methodological limitations of their era, established foundational principles that continue to inform current scholarship. Successive generations of researchers have built upon these foundations, progressively refining theoretical models and expanding the empirical evidence base through increasingly sophisticated research designs.

C Contemporary research has identified several critical factors that determine outcomes in this domain. The interplay between environmental conditions, technological capabilities, and social organisation creates a complex adaptive system in which small changes can produce disproportionately large effects. Understanding these dynamics requires analytical approaches that move beyond simple cause-and-effect reasoning to embrace the principles of complexity science and systems thinking.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F Looking ahead, the field stands at a critical juncture. Emerging technologies offer unprecedented opportunities for data collection and analysis, potentially enabling researchers to address questions that have long remained beyond the reach of empirical investigation. At the same time, the accelerating pace of social and technological change means that the phenomena under study are themselves evolving, requiring theoretical frameworks that can accommodate dynamism and discontinuity rather than assuming stability and equilibrium.

G In conclusion, the study of the melting of arctic sea ice represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Question 1

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 1 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 2

Choose the correct letter, A, B, C or D.

- 2 What finding emerged from the landmark study discussed in the passage?
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications

Questions 3–4

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 3 Early researchers had access to the same technologies as modern scholars.
- 4 The landmark study discussed in the passage involved participants from multiple countries.

Questions 5–6

Choose the correct letter, A, B, C or D.

- 5 According to the passage, what is one of the key factors identified by researchers?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations
- 6 The author suggests that early studies in this field were limited by:
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Question 7

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 7 The passage identifies power _____ as a key factor at the organisational level.

Question 8

Choose the correct letter, A, B, C or D.

- 8 What finding emerged from the landmark study discussed in the passage?
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Questions 9–11

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 9 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.
- 10 The author notes that researchers have called for more _____ approaches to investigation.
- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Question 12

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 12 The landmark study discussed in the passage involved participants from multiple countries.

Question 13

Choose the correct letter, A, B, C or D.

- 13 According to the passage, what is one of the key factors identified by researchers?
- A. minimal impact
 - B. creating underlying conditions for action
 - C. determining individual choices directly
 - D. irrelevant to outcomes